Synergy Klingon — Product Specification & Build Plan

1. CORE PHILOSOPHY: THE SYNERGY METHOD

1.1 What is the Synergy Method?

The Synergy method (inspired by Marcus Santamaria's Synergy Spanish) teaches languages through **combinatorial building**, not memorization. The learner acquires a small set of high-utility words divided into two categories:

- **Bricks**: High-frequency content words (verbs, nouns, adjectives) that carry meaning.
- **Mortar**: Structural/functional pieces (prefixes, suffixes, connectors, particles) that glue bricks together into sentences.

The key insight: learning ~80 carefully chosen pieces lets you **build 1,000+ sentences** through combination, rather than memorizing 1,000 sentences individually.

1.2 Synergy Principles (enforce these throughout the app)

1. **Combination over memorization.** The learner never memorizes full sentences. They learn parts and practice *assembling* them. Every exercise should reinforce the combinatorial nature of the system.
2. **Speak from day one.** Audio pronunciation is mandatory at every stage. The learner hears every word, every sentence, and is encouraged to speak along. There is no silent study phase.
3. **80/20 vocabulary selection.** Only the most common, most versatile everyday conversational words are included. No war vocabulary, no obscure terms. Every word must justify its place by the number of useful sentences it enables.
4. **Scaffolded complexity.** Each module introduces new bricks and mortar that combine with ALL previously learned pieces. Module 3 vocabulary works with Module 1 mortar. The learner's "sentence space" grows exponentially, not linearly.
5. **Immediate application.** Every new piece is immediately practiced in context — first in isolation (flashcard), then in guided combination (sentence builder), then in free recall (quiz), then in open practice (sandbox).

1.3 Why Klingon Fits Perfectly

Klingon (tlhIngan Hol) is agglutinative — it literally works by snapping pieces together:

- **Verb prefixes** encode subject AND object in one syllable (e.g., qa- = "I do something to you"). Learning 8 prefixes multiplies every verb ×8 instantly.
- **Verb suffixes** stack in defined slots: -be' (not), -qu' (very), -taH (ongoing), -laH (can), -pu' (completed), -'a' (question). These multiply every sentence further.
- **Noun suffixes** snap on for possession (-wIj my, -lIj your) and location (-Daq at/to).

- **Adjectives are verbs**, so they follow the same rules — no separate grammar track needed.

This means the Synergy "snap together" metaphor isn't just pedagogical — it's how the language actually works.

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2. COURSE CONTENT & LEARNING PATH

2.1 Module Structure

Each module follows a strict 4-phase learning sequence:

```
LEARN → BUILD → QUIZ → SANDBOX
(know the pieces) → (guided assembly) → (test recall) → (free practice)
```

Phase details:

Phase 1: LEARN (Flashcards)

- Present each new brick and mortar as a flashcard.
- Front: Klingon word in large text + speaker button that pronounces it.
- Back (on tap/click): English meaning, word type (verb/noun/prefix/suffix), and a usage hint.
- After revealing, user rates: "Got it" or "Need practice" (this feeds the spaced repetition system — see §3.4).
- All previously learned words from earlier modules can optionally appear as review cards.
- Completion: User must flip through all cards. Awards 15 XP.

Phase 2: BUILD (Sentence Constructor) — THE CORE EXERCISE

- User is shown an English sentence.
- Below it, a set of Klingon word/morpheme tiles are displayed in shuffled order.
- User taps tiles to construct the Klingon sentence in the correct order.
- Tiles must handle both **spacing and agglutination**:
  - Some pieces attach directly (prefix `jI` + verb `Sop` → `jISop`, suffix `-Daq` attaches to noun).
  - Some pieces are separate words (`bIQ vItlhutlh` — two words).
  - The tile data must specify whether each tile attaches to the previous/next tile or stands alone (see §4.2 for data model).
- On submit: correct answer plays success sound + shows the full Klingon sentence with pronunciation button. Wrong answer shows the correct answer, plays error sound, and the sentence is re-queued to appear again later in the set.

- Streak counter: consecutive correct answers increase XP multiplier.
- Each build exercise also has a "hear the whole sentence" button after completion.
- Completion: All sentences answered correctly (including re-queued ones). Awards 10 XP per sentence + streak bonuses.

Phase 3: QUIZ (Multiple Choice + Reverse Translation)

- Mix of question types:
  - **Klingon → English**: "What does jIyajbe' mean?" with 4 English options.
  - **English → Klingon**: "How do you say 'I eat'?" with 4 Klingon options.
  - **Audio recognition**: Play a Klingon word/sentence, user picks the English meaning.
  - **Fill the gap**: "I ___ water" → pick the correct Klingon verb.
- Each question has a "hear it" button on any Klingon text.
- 8-10 questions per quiz, drawn from current module + review from previous modules.
- Scoring: 15 XP per correct + streak bonuses. Must score ≥70% to pass and unlock next module.
- Failed quiz can be retaken immediately.

Phase 4: SANDBOX (Free Sentence Builder) — unlocked after quiz

- User has access to ALL learned bricks and mortar from all completed modules.
- Open canvas: user drags/taps pieces to build any sentence they want.
- A "Check" button validates whether their construction is grammatically plausible.
- A "Hear it" button pronounces whatever they've built.
- A "Translate" button shows the English meaning.
- This phase has no score or completion — it's a playground. But using it awards small XP (2 XP per valid sentence built, max 20 XP per session).
- Display a "Sentences built" counter and a "Unique combinations" counter to reinforce the Synergy philosophy ("Look — you know 47 pieces but you've built 200+ sentences!").

2.2 The Five Modules

MODULE 1: Survival Basics — "Eat, Drink & Get Around"

Theme: Immediate physical needs and navigation.

Bricks (13 words):

Klingon	English	Type
Sop	eat	verb
tlhutlh	drink	verb
jaH	go	verb
ghoS	come, approach	verb
ba'	sit	verb
Qong	sleep	verb
ghop	grab, take	verb
Soj	food	noun
bIQ	water	noun
HIq	ale, alcohol	noun
Qe'	restaurant	noun
juH	home	noun
pa'	room	noun

Mortar (6 pieces):

Klingon	English	Type
jI-	I (do, no object)	verb prefix
bI-	you (do, no object)	verb prefix
ma-	we (do, no object)	verb prefix
vI-	I → it/him/her	verb prefix
-Daq	at, to, in (location)	noun suffix
-'a'	yes/no question marker	verb suffix

Synergy math: 7 verbs × 4 prefixes = 28 basic sentences. Add -'a' = 56. Add -Daq with 3 location nouns = 168+ combinations. Total buildable sentences: **~200+**

Build exercises (8-10 sentences): Examples:

- "I eat." → jI + Sop → jISop
- "I drink water." → bIQ + vI + tlhutlh → bIQ vItlhutlh
- "We go home." → juH + Daq + ma + jaH → juHDaq majaH
- "I eat at the restaurant." → Qe' + Daq + jI + Sop → Qe'Daq jISop
- "Do you sleep?" → bI + Qong + 'a' → bIQong'a'
- "Do you drink ale?" → HIq + Da + tlhutlh + 'a' → HIq Datlhutlh'a'
- "I sit in the room." → pa' + Daq + jI + ba' → pa'Daq jIba'
- "We drink water." → bIQ + wI + tlhutlh → bIQ wItlhutlh

Note: Da- (you → it) is introduced here via build exercises even though formally taught in Module 2, because it appears in common sentences. This is intentional — the Synergy method exposes patterns slightly before formal teaching to prime recognition. Include Da- and wI- (we → it) as "bonus mortar" in the build exercise tile pool with a tooltip "New! you→it" to flag it.

MODULE 2: Opinions & Feelings — "Like, Want & Think"

Theme: Expressing internal states and preferences.

Bricks (10 words):

Klingon	English	Type
parHa'	like (lit. "un-dislike")	verb
par	dislike	verb
neH	want	verb
Qub	think	verb
yaj	understand	verb
ghoj	learn	verb
QaQ	be good	verb/adj
qab	be bad	verb/adj
Daj	be interesting	verb/adj
chu'	be new	verb/adj

Mortar (5 pieces):

Klingon	English	Type
Da-	you → it/him/her	verb prefix
qa-	I → you	verb prefix
cho-	you → me	verb prefix
-qu'	very, emphasis	verb suffix
-be'	not, negation	verb suffix

Synergy math: Now the learner has 8 prefixes × 13 verbs = 104 base forms. Add -be', -qu', -'a' = 400+ variants. With nouns and -Daq: **~600+** buildable sentences.

Build exercises (8-10 sentences):

- "I like the food." → Soj + vI + parHa' → Soj vIparHa'
- "It's very good!" → QaQ + qu' → QaQqu'
- "I don't understand." → jI + yaj + be' → jIyajbe'
- "I want water." → bIQ + vI + neH → bIQ vIneH
- "Do you like ale?" → HIq + Da + parHa' + 'a' → HIq DaparHa''a'
- "I understand you." → qa + yaj → qayaj
- "You understand me." → cho + yaj → choyaj
- "Klingon is interesting." → Daj + tlhIngan + Hol → Daj tlhIngan Hol
- "I don't dislike the food." → Soj + vI + par + be' → Soj vIparbe'

MODULE 3: Social Life — "Talk, Meet & Connect"

Theme: Interacting with people, names, basic courtesy.

Bricks (10 words):

Klingon	English	Type
jatlh	speak, say	verb
ghel	ask	verb
jang	answer, reply	verb
nob	give	verb
ghom	meet, encounter	verb
tlho'	thank	verb
jup	friend	noun
puq	child	noun
pong	name	noun
Hol	language	noun

Mortar (3 pieces):

Klingon	English	Type
-wIj	my	noun suffix
-lIj	your	noun suffix
-Daj	his/her/its	noun suffix

Synergy math: 8 prefixes × 19 verbs = 152 base. With 6 suffixes, 10 nouns, 3 possessives, location suffix: **~900+** buildable sentences.

Build exercises:

- "I speak Klingon." → tlhIngan + Hol + vI + jatlh → tlhIngan Hol vIjatlh
- "Thank you." → qa + tlho' → qatlho'
- "What is your name?" → nuq + pong + lIj → nuq ponglIj
- "I meet your friend." → jup + lIj + vI + ghom → juplIj vIghom
- "My child learns." → ghoj + puq + wIj → ghoj puqwIj
- "I give food." → Soj + vI + nob → Soj vInob

Also introduce tlhIngan as a fixed compound brick meaning "Klingon (person/thing)" since it appears constantly.

MODULE 4: Time & Questions — "When, Where & How"

Theme: Temporal context and interrogatives.

Bricks (10 words):

Klingon	English	Type
DaH	now	time word
wa'leS	tomorrow	time word
wa'Hu'	yesterday	time word
tugh	soon	time word
reH	always	time word
not	never	time word
nuq	what?	question word
nuqDaq	where?	question word
chay'	how?	question word
qatlh	why?	question word

Mortar (3 pieces):

Klingon	English	Type
-pu'	completed action (perfective)	verb suffix
-taH	ongoing/continuous	verb suffix
-laH	can, able to	verb suffix

Synergy math: With time words prefixing any sentence and 3 new verb suffixes stacking: **~1,200+** buildable sentences.

Build exercises:

- "I am eating now." → DaH + jI + Sop + taH → DaH jISoptaH

- "Can you speak Klingon?" → tlhIngan + Hol + Da + jatlh + laH + 'a' → tlhIngan Hol DajatlhlaH'a'
- "Yesterday I drank ale." → wa'Hu' + HIq + vI + tlhutlh + pu' → wa'Hu' HIq vItlhutlhpu'
- "Where is your home?" → nuqDaq + juH + lIj → nuqDaq juHlIj
- "How do I learn?" → chay' + jI + ghoj → chay' jIghoj
- "I always want good food." → reH + Soj + QaQ + vI + neH → reH Soj QaQ vIneH

MODULE 5: Personality & Flow — "Express Yourself"

Theme: Emotions, courtesy phrases, commands, social fluency.

Bricks (9 words):

Klingon	English	Type
Hagh	laugh	verb
bel	be pleased	verb
tIv	enjoy	verb
nuqneH	What do you want? (greeting)	fixed phrase
Qapla'	Success! (farewell/toast)	fixed phrase
HISlaH	Yes	fixed phrase
ghobe'	No	fixed phrase
lu'	OK, agreed	fixed phrase
majQa'	Well done! Bravo!	fixed phrase

Mortar (2 pieces):

Klingon	English	Type
yI-	imperative/command prefix (you-do-it!)	verb prefix
-Qo'	don't! (negative imperative)	verb suffix

Synergy math: Full inventory of ~80 pieces. Buildable sentences: **well over 1,500.**

Build exercises:

- "I enjoy the food." → Soj + vI + tIv → Soj vItIv
- "We are laughing!" → ma + Hagh + taH → maHaghtaH
- "I am pleased." → jI + bel + taH → jIbeltaH
- "Are you enjoying?" → bI + tIv + 'a' → bItIv'a'
- "Sit down!" → yI + ba' → yIba'
- "Don't sleep!" → yI + Qong + Qo' → yIQongQo'
- "Speak! (command)" → yI + jatlh → yIjatlh

2.3 Cumulative Review System

- After completing Module 3, the quiz phase should include 20% review questions from Modules 1-2.
- After Module 4, quizzes include 25% review from all prior modules.
- After Module 5, a special "Master Review" quiz of 20 questions spanning all modules unlocks. Passing this (≥80%) awards a special rank.

2.4 Pronunciation Guide (Always Accessible)

A collapsible pronunciation reference panel available from any screen:

Symbol	Pronunciation	Audio Example
tlh	Place tongue on roof of mouth, release air from sides (like a lateral affricate)	Button
Q	Deep, harsh K from very back of throat (uvular)	Button
q	K from back of throat (further back than English K)	Button
gh	Gargling/voiced velar fricative	Button
H	Harsh, breathy H like clearing your throat	Button
ng	Like "ng" in "singing"	Button
'	Glottal stop — brief catch in the throat (like "uh-oh")	Button
I (capital i)	Always "ee" sound	Button
u	Always "oo" sound	Button
e	Always "eh" sound	Button
a	Always "ah" sound	Button
o	Always "oh" sound	Button

3. AUDIO & PRONUNCIATION SYSTEM

3.1 Strategy

There is no official Klingon text-to-speech engine. We use a hybrid approach:

Option A (preferred): Pre-recorded audio files

- Record (or source from existing Klingon language resources) audio clips for every word and sentence in the course.
- Store as `.mp3` or `.ogg` files, organized by module.
- File naming convention: `m1_sop.mp3`, `m1_sentence_jiSop.mp3`, etc.
- If professional recordings aren't available, use high-quality amateur recordings from the Klingon Language Institute community.

Option B (fallback): Web Speech API with phonetic mapping

- Create a phonetic dictionary mapping every Klingon word to its closest IPA/English phonetic approximation.
- Use the Web Speech API (`speechSynthesis`) with specific settings:
  - Rate: 0.7 (slower than normal for clarity)
  - Pitch: 0.8 (slightly deeper)
  - Preferred voice: a male, low-pitched voice if available
- This is an approximation — the phonetic mappings won't be perfect but give the learner a reasonable starting point.
- Phonetic dictionary must cover: every individual brick, every individual mortar piece, every complete build exercise answer sentence, and every quiz option that's in Klingon.

Option C (ideal future): Custom TTS

- Train a lightweight TTS model on available Klingon audio corpora.
- This is out of scope for v1 but should be architecturally planned for (abstract the audio playback behind an interface so the source can be swapped).

3.2 Phonetic Dictionary (for Option B)

Every Klingon word in the course needs a phonetic mapping. Here is the format:

```
json
```

{
"Sop": "SHOP",
"tlhutlh": "TLOOTL",
"jaH": "JAH",
"ghoS": "GHOSH",
"ba'": "BAH",
"Qong": "KHONG",
"Soj": "SHOJ",
"bIQ": "BIKH",
"HIq": "HEEKH",
"Qe'": "KHEH",
"juH": "JOOH",
"pa'": "PAH",
"parHa'": "PAR-HAH",
"par": "PAR",
"neH": "NEH",
"Qub": "KHOOB",
"yaj": "YAZH",
"ghoj": "GHOJ",
"QaQ": "KHAKH",
"qab": "KHAB",
"Daj": "DAJ",
"chu'": "CHOO",
"jatlh": "JATL",
"ghel": "GHEL",
"jang": "JANG",
"nob": "NOB",
"ghom": "GHOM",
"tlho'": "TLOW",
"jup": "JOOP",
"puq": "POOKH",
"pong": "PONG",
"Hol": "HHOL",
"DaH": "DAH",
"wa'leS": "WAH-LESH",
"wa'Hu'": "WAH-HOOH",
"tugh": "TOOGH",
"not": "NOT",
"reH": "REH",
"nuq": "NOOKH",
"nuqDaq": "NOOKH-DAKH",
"chay'": "CHAY",
"qatlh": "KHATL",
"Hagh": "HAHGH",
"bel": "BEL",
"tIv": "TEEV",

```
  "nuqneH": "NOOKH-NEH",
  "Qapla'": "KHAP-LAH",
  "HISlaH": "HHEESH-LAH",
  "ghobe'": "GHOH-BEH",
  "lu'": "LOO",
  "majQa'": "MAJ-KHAH"
}
```

Every build-exercise answer sentence must also have a full phonetic entry. Generate these by concatenating word phonetics with natural pauses (commas in the phonetic string) between words. Example:

- `jISop` → `"JEE-SHOP"`
- `bIQ vItlhutlh` → `"BIKH, VEE-TLOOTL"`
- `Qe'Daq jISop` → `"KHEH-DAKH, JEE-SHOP"`

3.3 Where Audio Plays

Speaker buttons (🔊 icon) must appear in these locations:

1. **Learn phase**: On every flashcard (front side — user can hear before flipping).
2. **Build phase**: On the English prompt (optional, for reverse listening), and on the answer reveal (mandatory — auto-plays on correct answer).
3. **Quiz phase**: On every Klingon option, and on audio-recognition questions (auto-plays the question).
4. **Sandbox**: "Hear it" button that pronounces whatever the user has built.
5. **Pronunciation guide**: On every sound example.

3.4 Sound Effects (Non-Speech)

Use Tone.js for UI sound effects. Keep these distinct from speech audio:

- **Card flip**: Short sine blip (D5, 32nd note)
- **Tile tap**: Gentle click (A4, 32nd note)
- **Correct answer**: Rising triad (C5 → E5 → G5, triangle wave)
- **Wrong answer**: Descending minor (E3 → Eb3, sawtooth)
- **Phase complete**: Resolved chord (C4+E4+G4+C5, triangle, sustained)
- **Rank up**: Ascending fanfare (C5 → E5 → G5 → C6, triangle, with slight delay between notes)
- **XP gain**: Quick sparkle (high sine, very short)
- A global mute/unmute toggle in the header for all sounds.

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4. DATA MODEL

4.1 Word/Morpheme Model

```typescript
interface KlingonPiece {
  id: string;              // unique ID, e.g. "v_sop", "pfx_ji", "sfx_daq"
  tlh: string;             // Klingon text as displayed
  en: string;              // English meaning
  type: "verb" | "noun" | "adj" | "prefix" | "suffix" | "time" | "question" | "phrase";
  phonetic: string;        // phonetic pronunciation string
  audioFile?: string;      // path to audio file (Option A)
  moduleIntroduced: number; // which module introduces this piece
  category: "brick" | "mortar";
  attachBehavior: "standalone" | "attach-before" | "attach-after";
  // attach-before: this piece attaches to the following piece (prefixes)
  // attach-after: this piece attaches to the preceding piece (suffixes)
  // standalone: this is a separate word
}
```

4.2 Build Exercise Model

```typescript
interface BuildExercise {
  id: string;
  moduleId: number;
  englishPrompt: string;     // "I eat at the restaurant."
  tiles: BuildTile[];        // the pieces the user must arrange
  correctAnswer: string;     // "Qe'Daq jISop" — the expected output string
  phonetic: string;          // "KHEH-DAKH, JEE-SHOP"
  audioFile?: string;
  explanation?: string;      // optional grammar note shown after correct answer
}

interface BuildTile {
  text: string;              // display text of the tile, e.g. "jI", "Sop", "Daq"
  attachRight: boolean;      // does this tile attach to the next tile (no space)?
  attachLeft: boolean;       // does this tile attach to the previous tile (no space)?
}
```

Critical implementation detail: When the user arranges tiles, the display must dynamically show them either attached (no space) or separated (space) based on the attach properties. For example, if tile `jI` has `attachRight: true` and tile `Sop` has `attachLeft: true`, they render as `jISop`. But `bIQ` with `attachRight: false` followed by `vI` renders as `bIQ vI...`.

4.3 Quiz Question Model

```typescript
interface QuizQuestion {
  id: string;
  moduleId: number;
  type: "tlh-to-en" | "en-to-tlh" | "audio-recognition" | "fill-gap";
  question: string;         // the question text or audio reference
  options: string[];        // 4 options
  correctIndex: number;     // index of correct option
  phonetic?: string;        // phonetic for any Klingon in the question
}
```

4.4 User Progress Model

```typescript
interface UserProgress {
  xp: number;
  currentRank: number;
  modulesCompleted: Record<string, boolean>; // "1-learn", "1-build", etc.
  streak: number;
  sentencesBuilt: number;       // lifetime count for sandbox
  uniqueSentencesBuilt: number; // unique combinations in sandbox
  wordConfidence: Record<string, number>; // word ID → confidence 0-5 for spaced repetition
  quizScores: Record<number, number>;     // moduleId → best percentage
}
```

4.5 Rank System

```typescript
const ranks = [
  { xp: 0,   tlh: "tlhIngan Hol chu'wI'", en: "Klingon Newcomer",   icon: "🌑" },
  { xp: 50,  tlh: "ghojwI'",              en: "Student",            icon: "🌒" },
  { xp: 150, tlh: "paQDI'norgh ghajwI'",  en: "Knowledge Holder",   icon: "🌓" },
  { xp: 300, tlh: "Hol pIn",              en: "Language Expert",    icon: "🌔" },
  { xp: 500, tlh: "Hol po'wI'",           en: "Language Master",    icon: "🌕" },
  { xp: 750, tlh: "Hol qoq",              en: "Language Legend",    icon: "🌟" },
];
```

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5. GAMIFICATION RULES

5.1 XP Awards

Action	Base XP	Streak Bonus
Complete Learn phase	15	—
Correct Build sentence	10	+2 per streak (max +10)
Correct Quiz answer	15	+3 per streak (max +15)
Sandbox valid sentence	2	— (max 20/session)
Complete a module (all 3 phases)	25 bonus	—
Pass Master Review	100	—

5.2 Streak System

- Streak increments on each consecutive correct answer within a phase (build or quiz).
- Streak resets to 0 on any wrong answer.
- Display streak counter with fire emoji (🔥) when streak ≥ 2.
- At streak milestones (5, 10), play a special sound and show a brief celebration animation.

5.3 Module Unlock Logic

- Module 1 is always unlocked.
- Module N (N > 1) unlocks when the user has completed the Quiz phase of Module N-1 with ≥70%.
- Sandbox for a module unlocks after its Quiz phase is passed.
- The Master Review unlocks after all 5 module quizzes are passed.

5.4 Spaced Repetition (Lightweight)

- Each word has a confidence score (0-5).
- Score starts at 0. Increments by 1 on correct recall, decrements by 1 on incorrect.
- Words with confidence < 3 appear more frequently in review cards and quiz questions.
- The Learn phase "Got it" / "Need practice" buttons set initial confidence (2 vs 0).
- This doesn't need a full SRS algorithm — a simple weighted random selection is fine for v1.

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6. UI/UX SPECIFICATIONS

6.1 Visual Design

- **Theme**: Dark space/sci-fi theme. Deep navy/purple gradients. Klingon aesthetic without being campy.
- **Color palette**:
  - Background: #0f0f1a → #1a1025 gradient
  - Module 1: #f97316 (orange)
  - Module 2: #a855f7 (purple)
  - Module 3: #06b6d4 (cyan)
  - Module 4: #eab308 (gold)
  - Module 5: #ec4899 (pink)
  - Success: #22c55e
  - Error: #ef4444
  - XP/Gold: #fbbf24
  - Text primary: #e2e8f0
  - Text secondary: #64748b
- **Typography**: System sans-serif for UI. Monospace for Klingon text (it makes the alien script feel distinct).
- **Klingon text should always be visually distinct** from English — larger, bolder, in a monospace or display font with slightly more letter-spacing.

6.2 Layout

- Mobile-first, max-width 520px centered. Must work beautifully on phones.
- Bottom navigation is optional but a header with XP display and sound toggle is required.
- Smooth transitions between views (fade or slide).
- No page reloads — single-page app with client-side routing.

6.3 Key Interactions

- Flashcards: Tap to flip (with flip animation — CSS 3D transform).
- Build tiles: Tap to select/deselect. Selected tiles animate into the answer zone. Tiles in the answer zone can be tapped to return them.
- Quiz options: Tap to select. After selection, correct answer highlights green, wrong highlights red, with a brief delay before the "Next" button appears.
- All interactive elements should have hover/active states.
- Speaker buttons: consistent 🔊 icon, should show a brief "playing" animation (pulse or sound wave).

6.4 Responsive Behavior

- On narrow screens (<360px), tiles in build exercises may need to wrap to 2 rows.
- Flashcards should fill available width minus padding.
- Font sizes should not go below 13px for readability.

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7. TECHNICAL ARCHITECTURE

7.1 Technology Stack (Recommended)

- **Framework**: React (functional components, hooks)
- **Styling**: Tailwind CSS or inline styles (for artifact compatibility)
- **Audio - Speech**: Web Speech API (SpeechSynthesis) with phonetic dictionary fallback, abstracted behind an `AudioService` interface for future swapping to real audio files
- **Audio - SFX**: Tone.js for all non-speech sounds
- **State Management**: React useState/useReducer (no external state library needed)
- **Persistence**: Use the `window.storage` API for progress persistence across sessions. Store user progress under key `synergy-klingon-progress`. Load on app init, save after every XP change or completion event.
- **Routing**: Simple state-based view switching (dashboard / lesson / sandbox)

7.2 Key Component Tree

```
App
├── Header (XP bar, rank, sound toggle)
├── Dashboard
│   ├── RankCard
│   ├── ModuleCard × 5
│   └── PronunciationGuide
├── LessonView
│   ├── PhaseNav (Learn | Build | Quiz | Sandbox)
│   ├── LearnPhase (flashcard stack)
│   ├── BuildPhase (sentence constructor)
│   ├── QuizPhase (question flow)
│   └── SandboxPhase (free builder)
├── MasterReview
└── RankUpOverlay (modal)
```

7.3 AudioService Interface

```typescript
```

```
interface AudioService {
  speakWord(klingonText: string): Promise<void>;
  speakSentence(klingonText: string): Promise<void>;
  playSFX(type: "correct" | "wrong" | "flip" | "click" | "complete" | "levelup" | "sparkle"): void;
  setMuted(muted: boolean): void;
  isMuted(): boolean;
}
```

Implement with Web Speech API + Tone.js. The `speakWord` and `speakSentence` methods look up the phonetic dictionary and pass the phonetic string to `speechSynthesis.speak()` with the voice settings from §3.1 Option B.

7.4 Persistence Strategy

On every significant state change (XP gain, phase completion, quiz score):

```javascript
await window.storage.set('synergy-klingon-progress', JSON.stringify(userProgress));
```

On app load:

```javascript
try {
  const saved = await window.storage.get('synergy-klingon-progress');
  if (saved) setProgress(JSON.parse(saved.value));
} catch { /* start fresh */ }
```

---

8. WHAT SUCCESS LOOKS LIKE

After completing all 5 modules and the Master Review, the user should be able to:

1. Construct basic conversational sentences in Klingon from memory.
2. Pronounce all words with reasonable accuracy.
3. Understand the OVS word order instinctively.
4. Use verb prefixes, verb suffixes, noun suffixes, time words, and question words in combination.
5. Feel confident they can express basic needs, opinions, social pleasantries, and questions.
6. Have internalized the Synergy principle: "I know 80 pieces and can say 1,500 things."

The app should feel rewarding, clear, and alive with sound — not like a textbook, but like a game that happens to teach you a language.

---

9. FUTURE ENHANCEMENTS (Out of scope for v1, but plan for them)

- Custom Klingon TTS model
- Voice recognition for pronunciation practice
- Daily challenge mode (random sentence building from full vocabulary)
- Social sharing ("I can say 847 things in Klingon!")
- Additional modules: Shopping, Weather, Family, Storytelling
- pIqaD script mode (Klingon native writing system)
- Conversation simulator (scripted dialogues with branching)